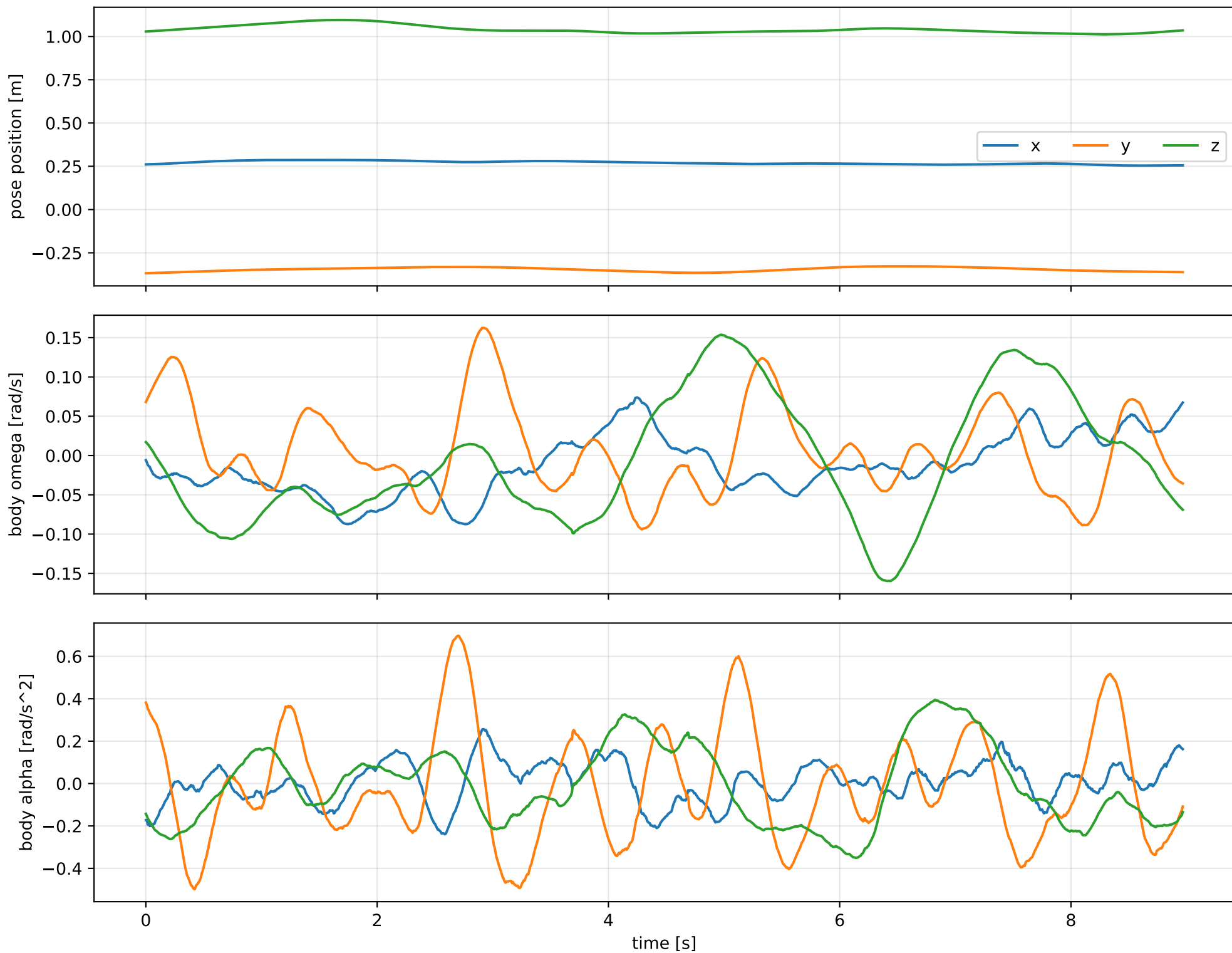
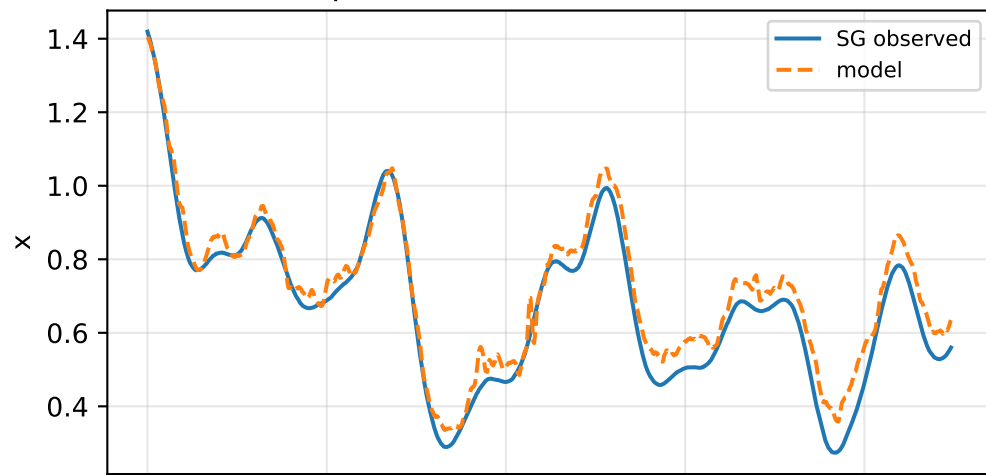


gimbal_measured_linear: SG trajectory and derived motion

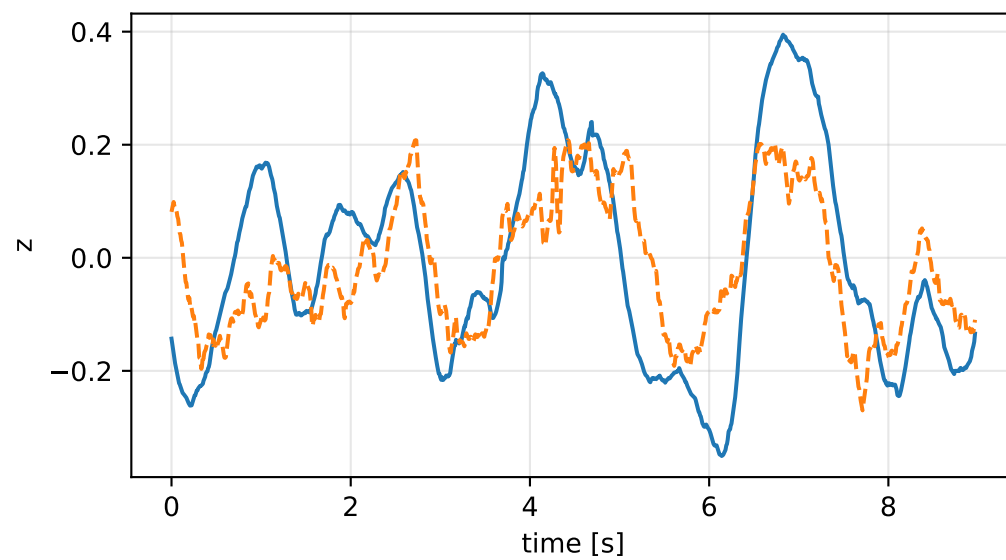
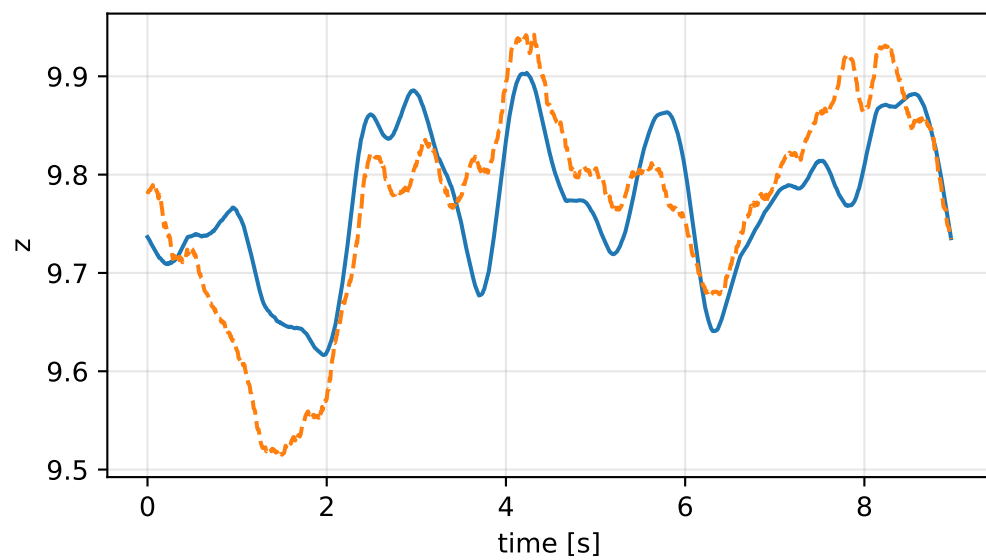
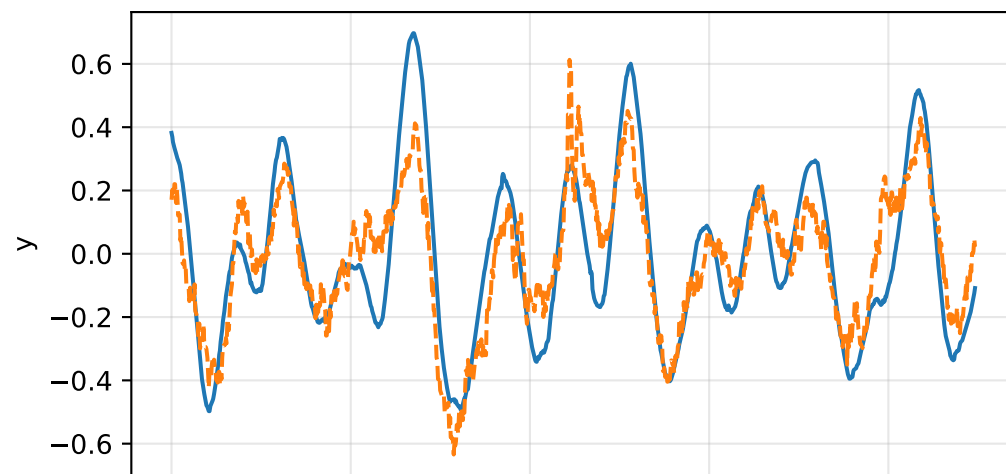
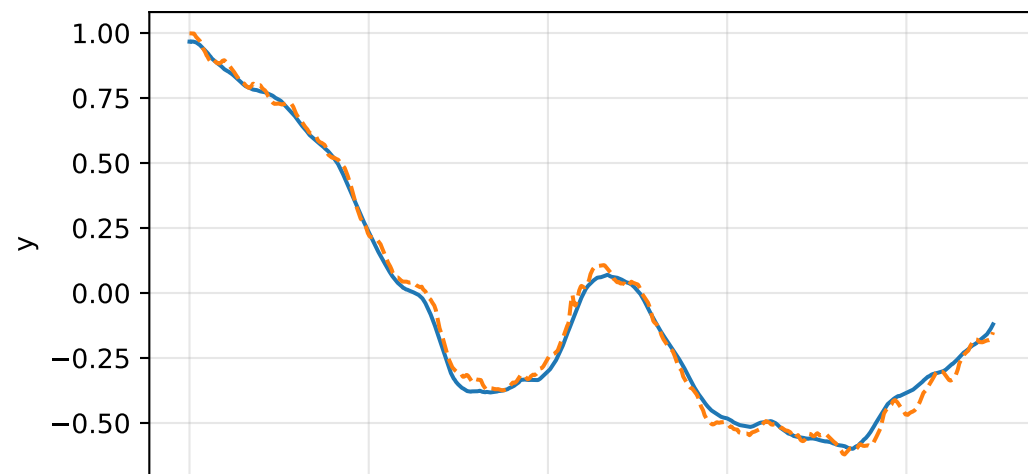
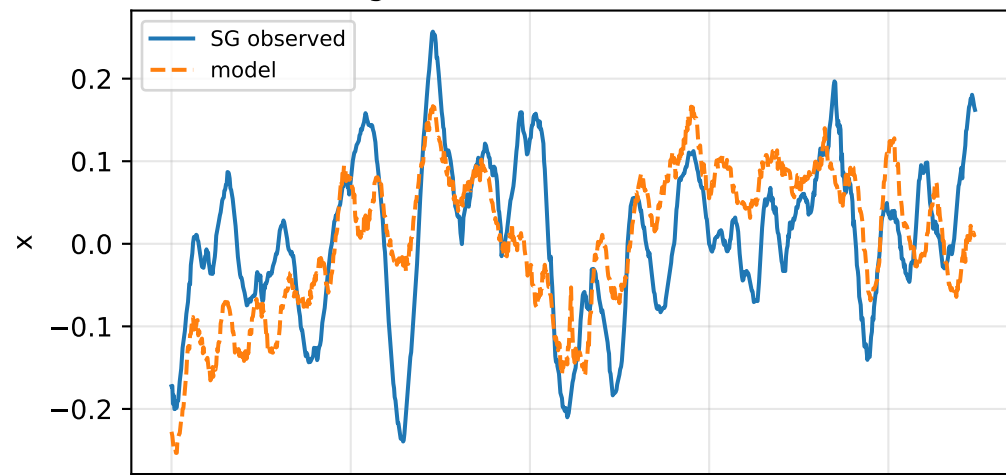


gimbal_measured_linear: acceleration residual objective

specific acceleration [m/s²]

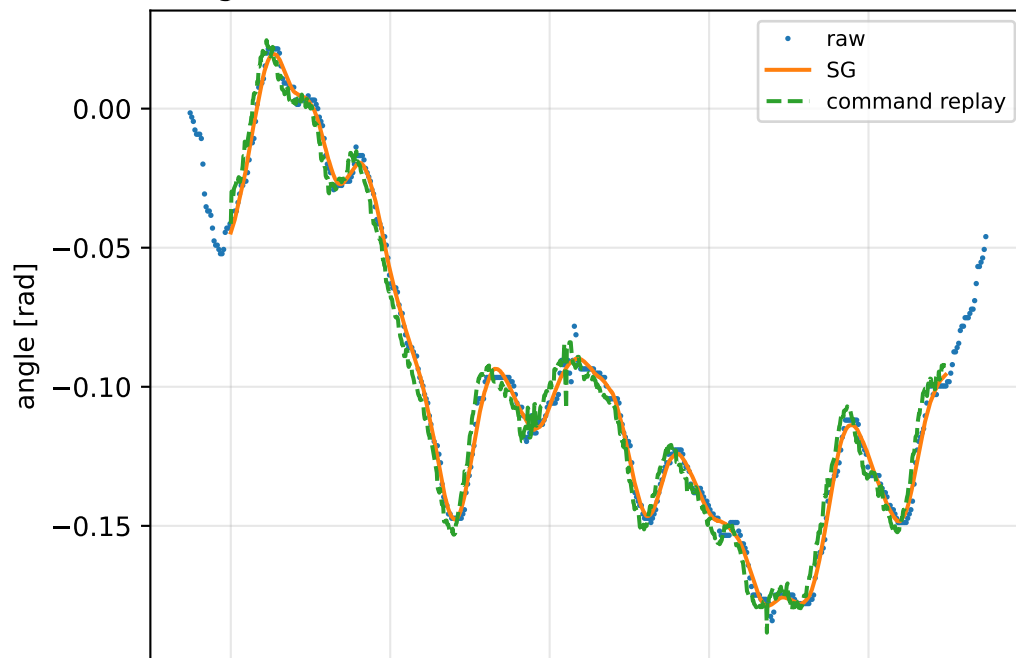


angular acceleration [rad/s²]

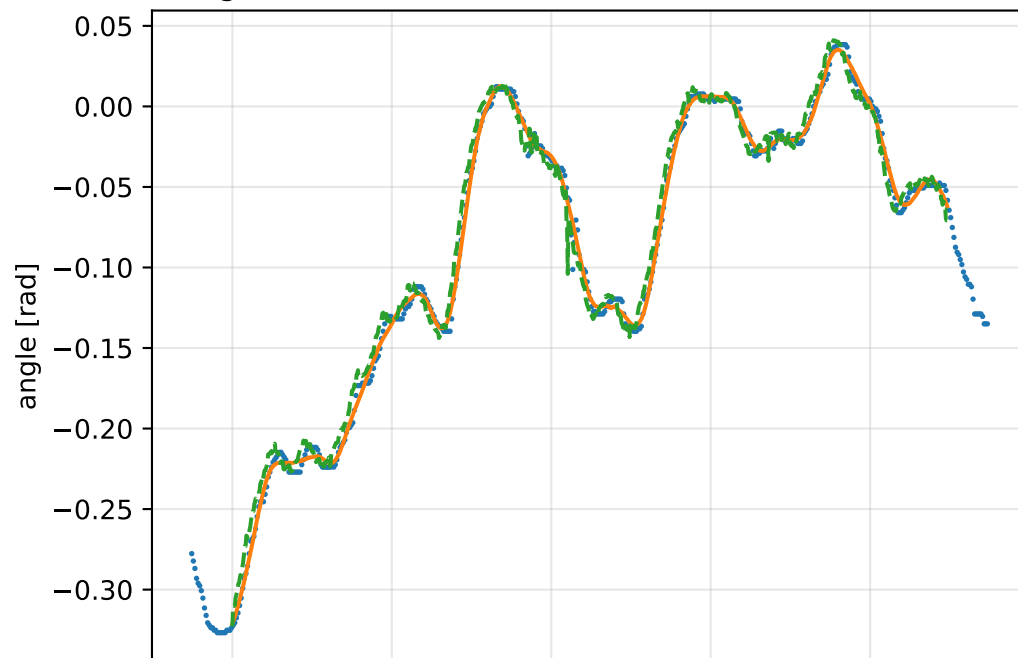


gimbal_measured_linear: gimbal measurement audit (objective=measured_linear)

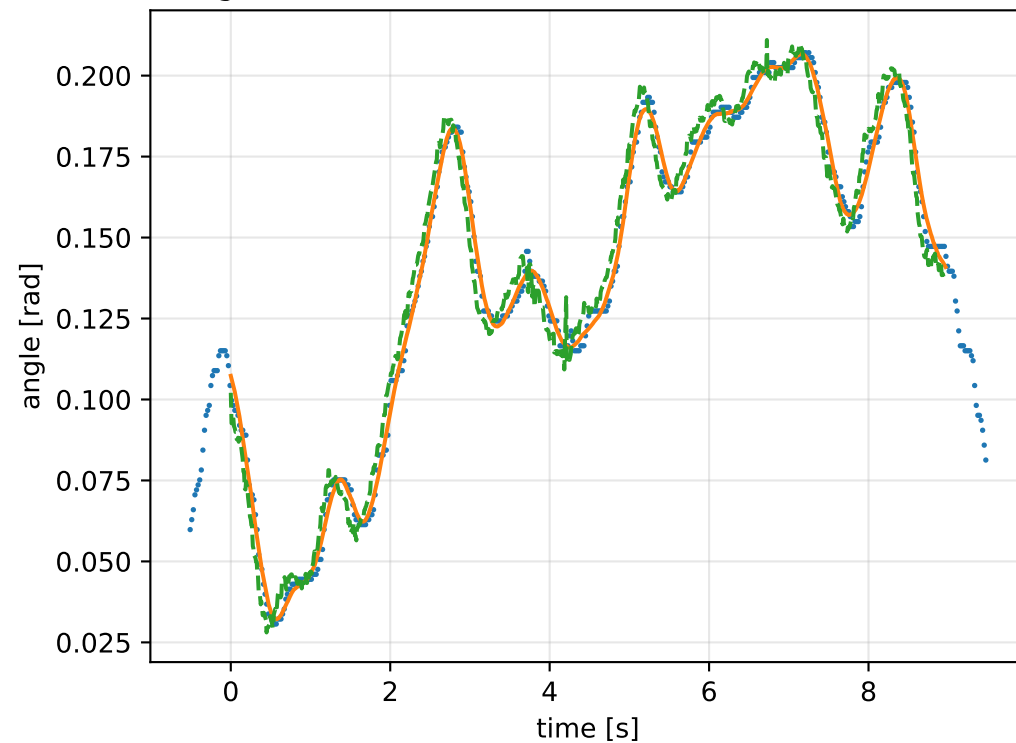
gimbal 1: RMSE=0.006345, max=0.01494 rad



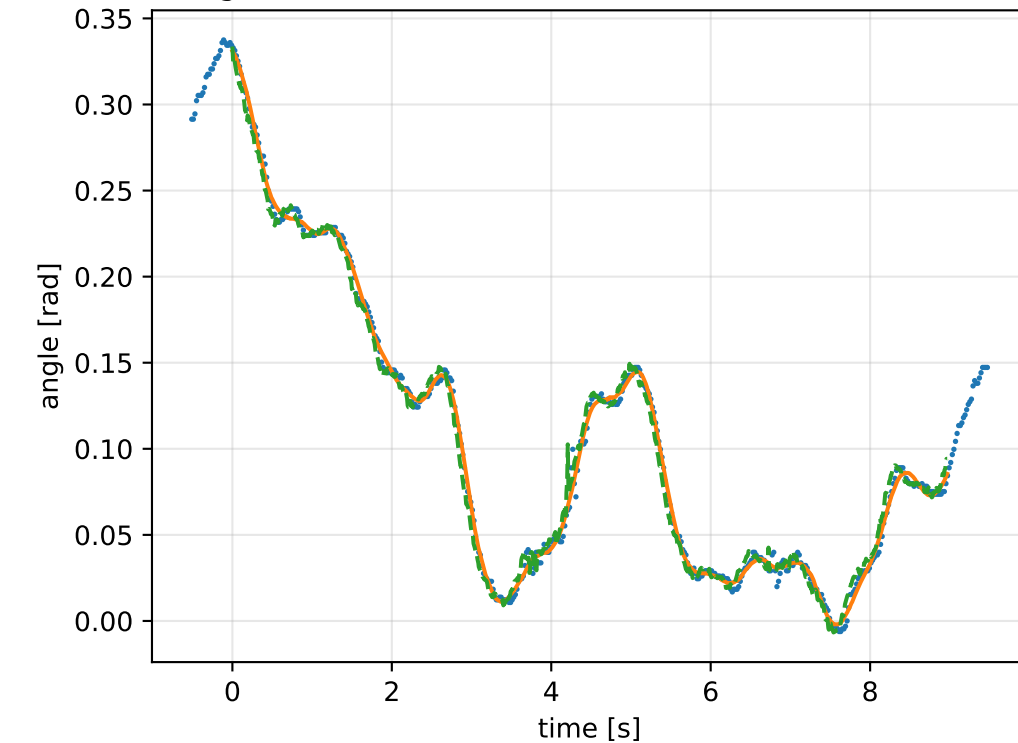
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

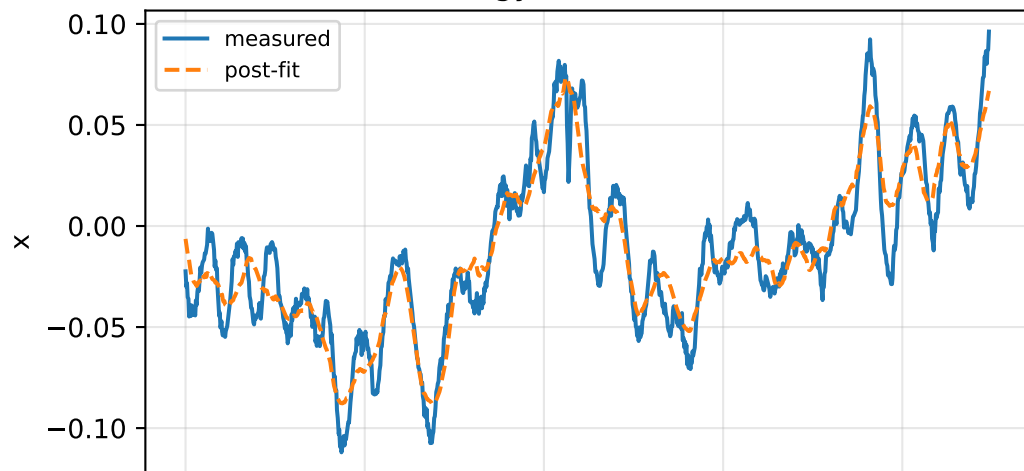


gimbal 4: RMSE=0.006321, max=0.03642 rad

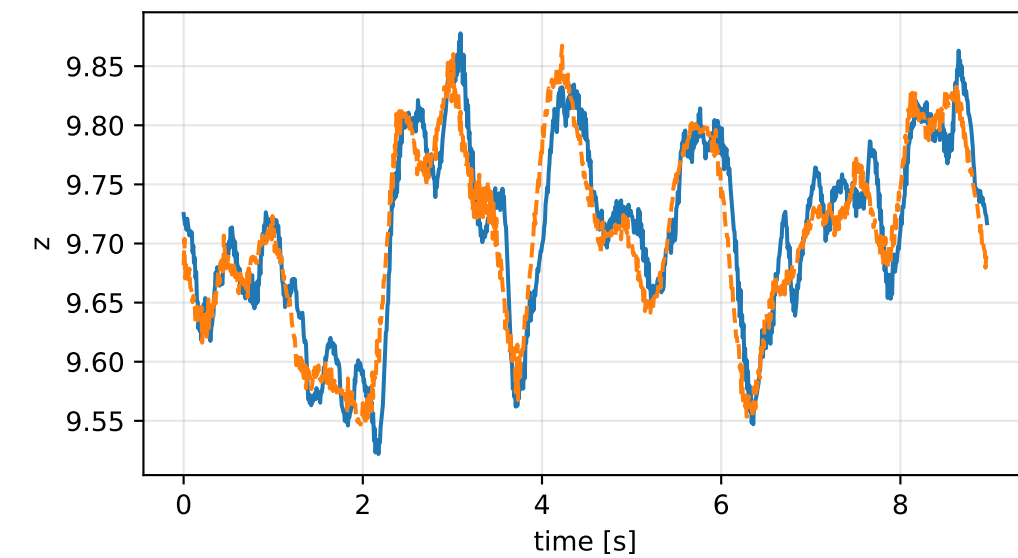
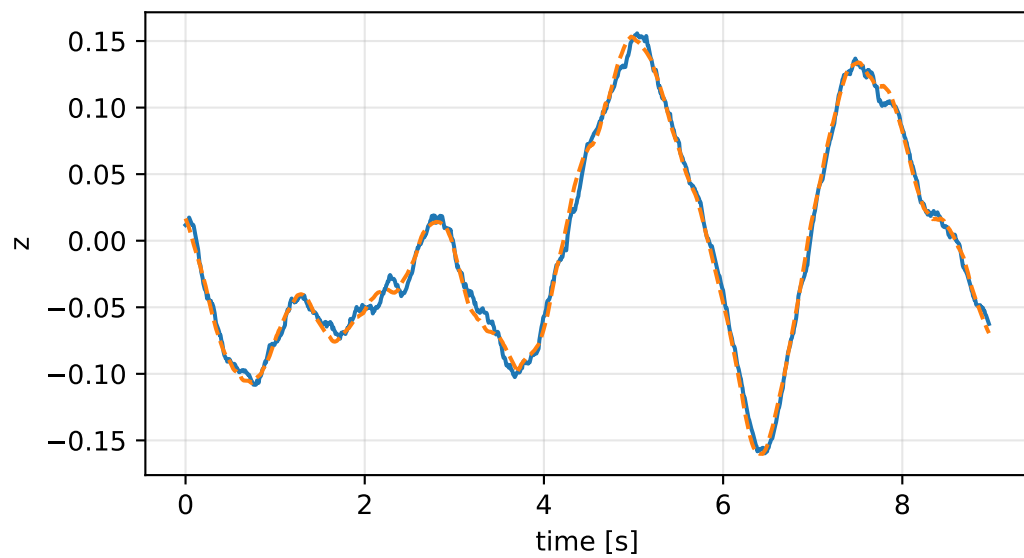
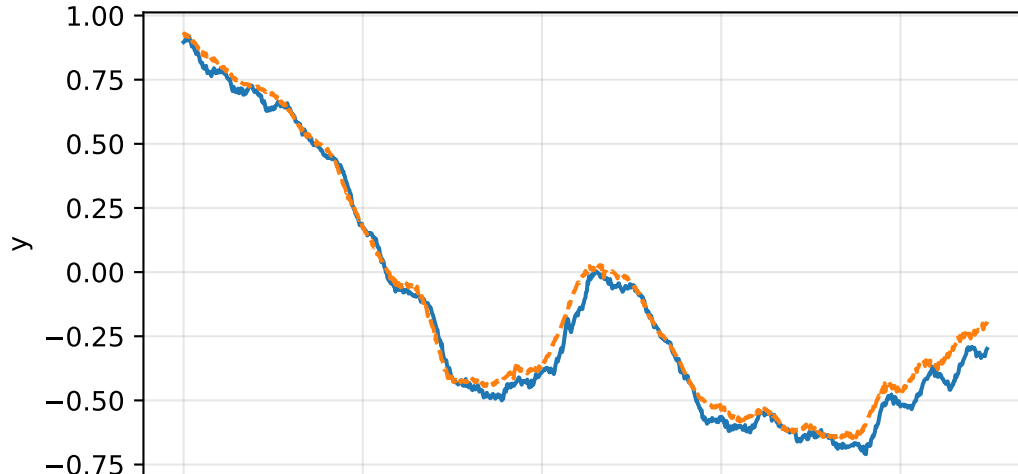
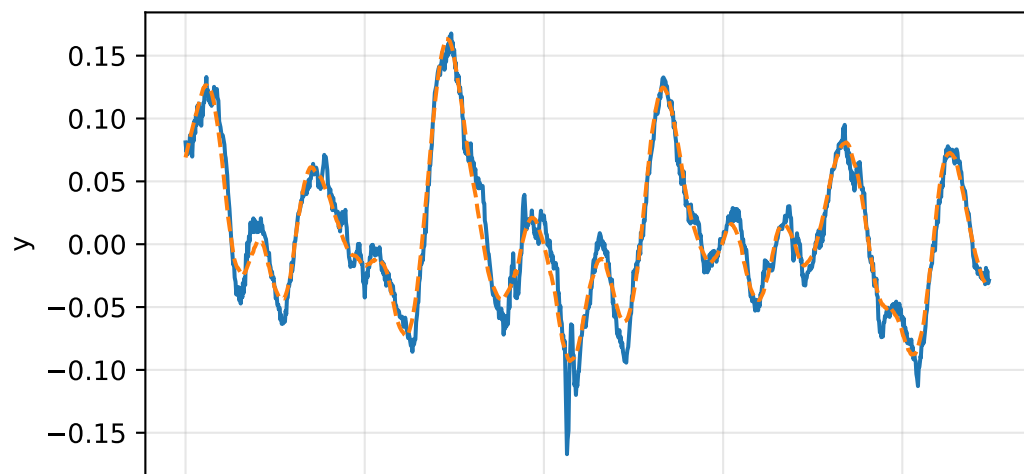
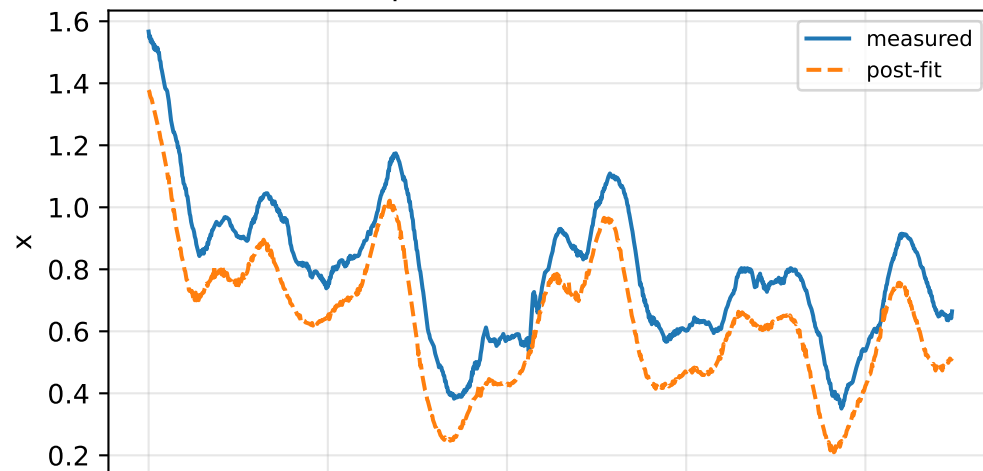


gimbal_measured_linear: IMU comparison (diagnostic only)

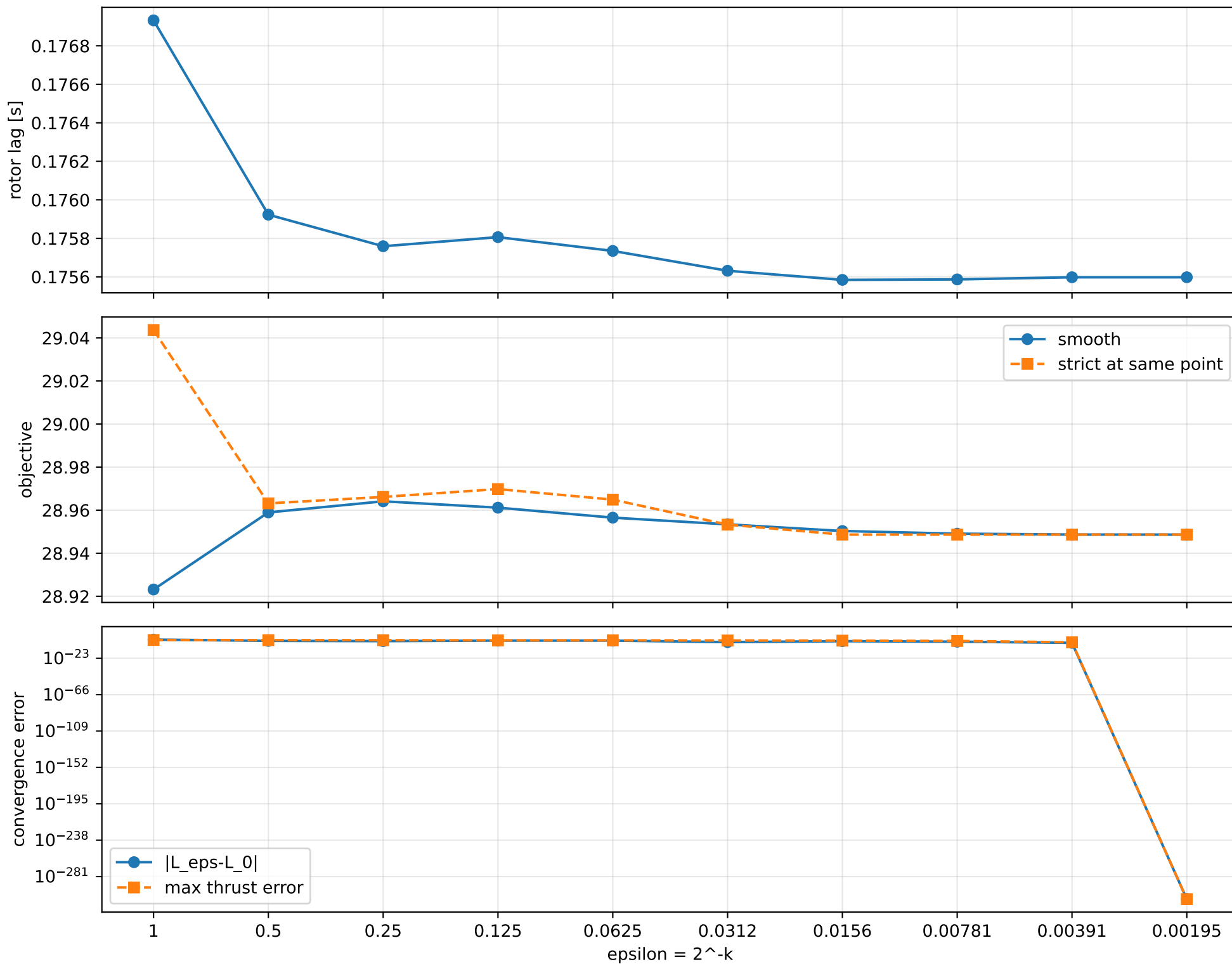
gyro [rad/s]



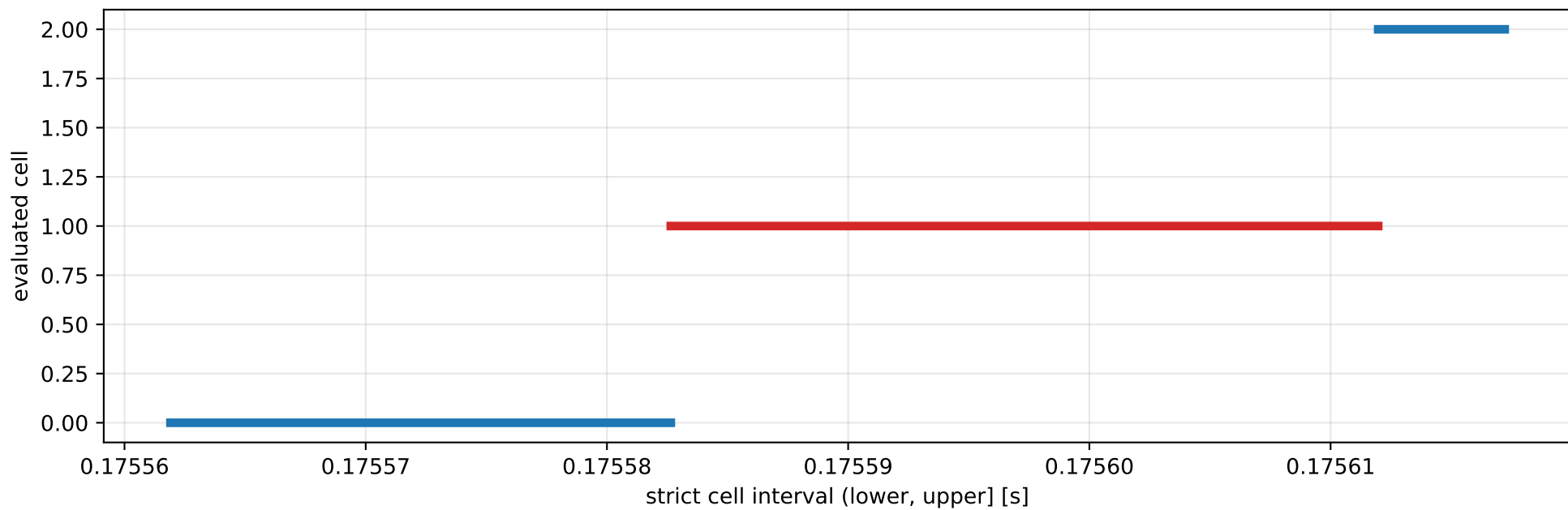
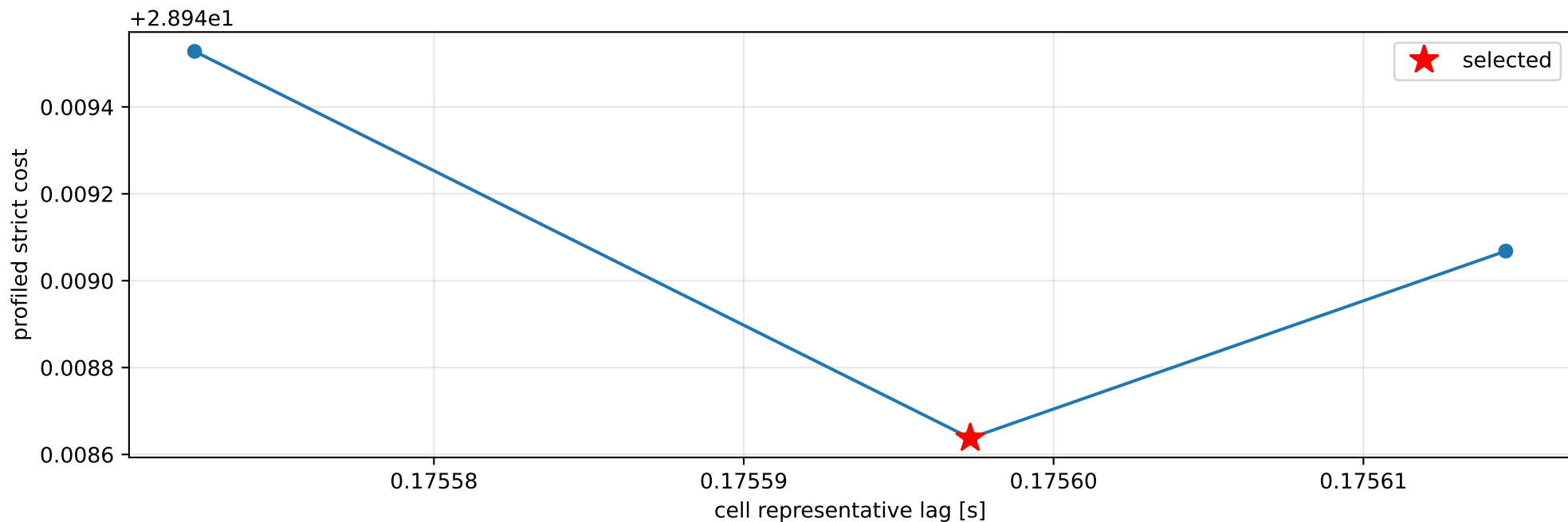
specific force [m/s^2]



gimbal_measured_linear: smooth-to-strict continuation



gimbal_measured_linear: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175597974 s; data boundary=False

gimbal_measured_linear: parameter estimate

Case: gimbal_measured_linear

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.24247777

inertia [kg m²]:

```
[[ 1.26340763 -0.02684186  0.09250722]
 [-0.02684186  0.57523664 -0.02090078]
 [ 0.09250722 -0.02090078  0.72141888]]
```

CoG body [m]: [-0.0202679 -0.05513659 -0.00579307]

force effectiveness: [2.33177398 2.2789935 1.1981018 1.46561446]

scale-free J/m [m²]:

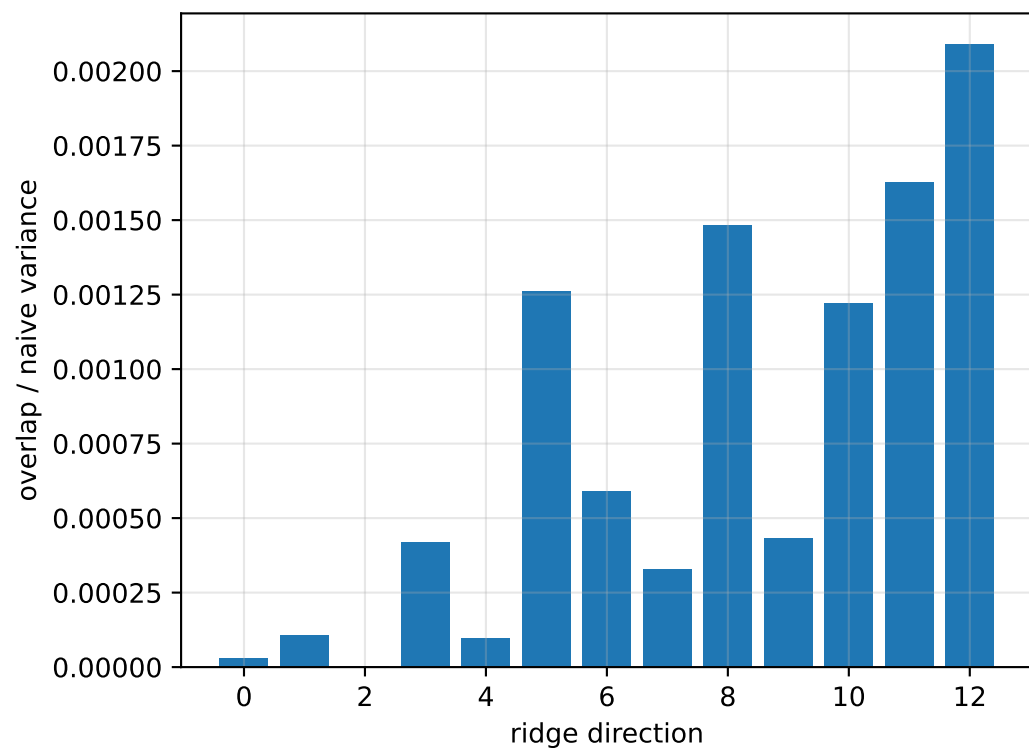
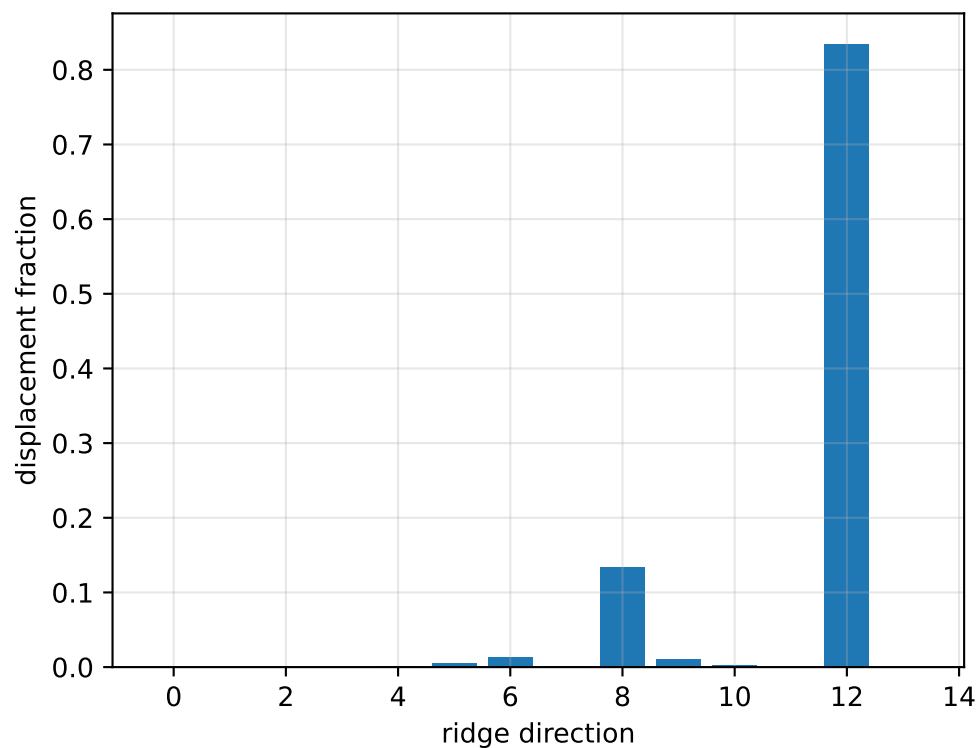
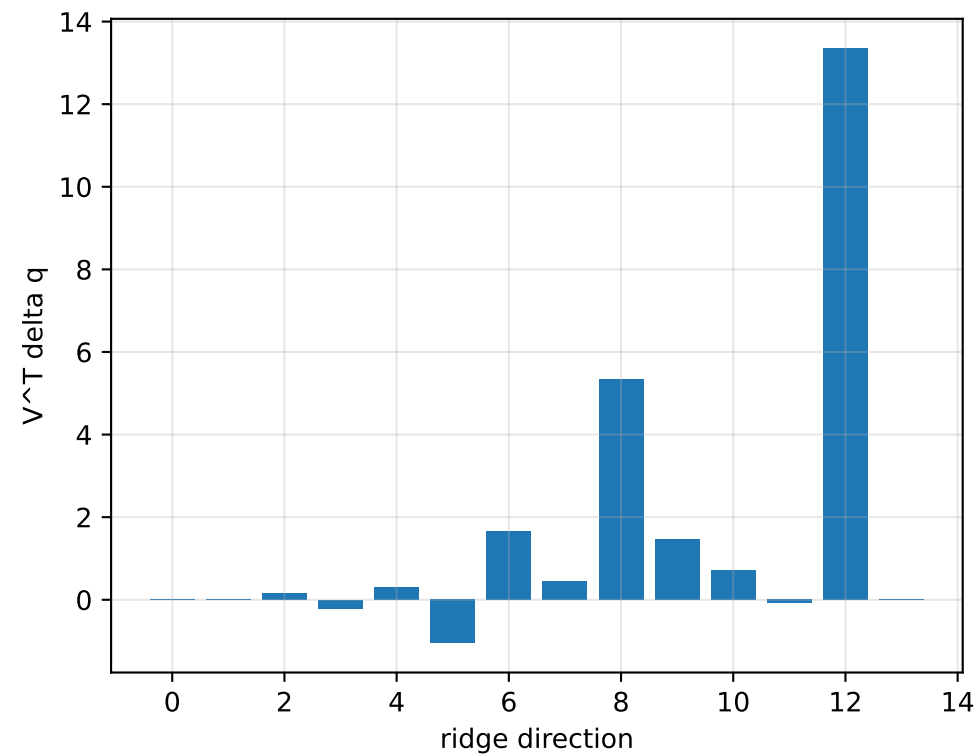
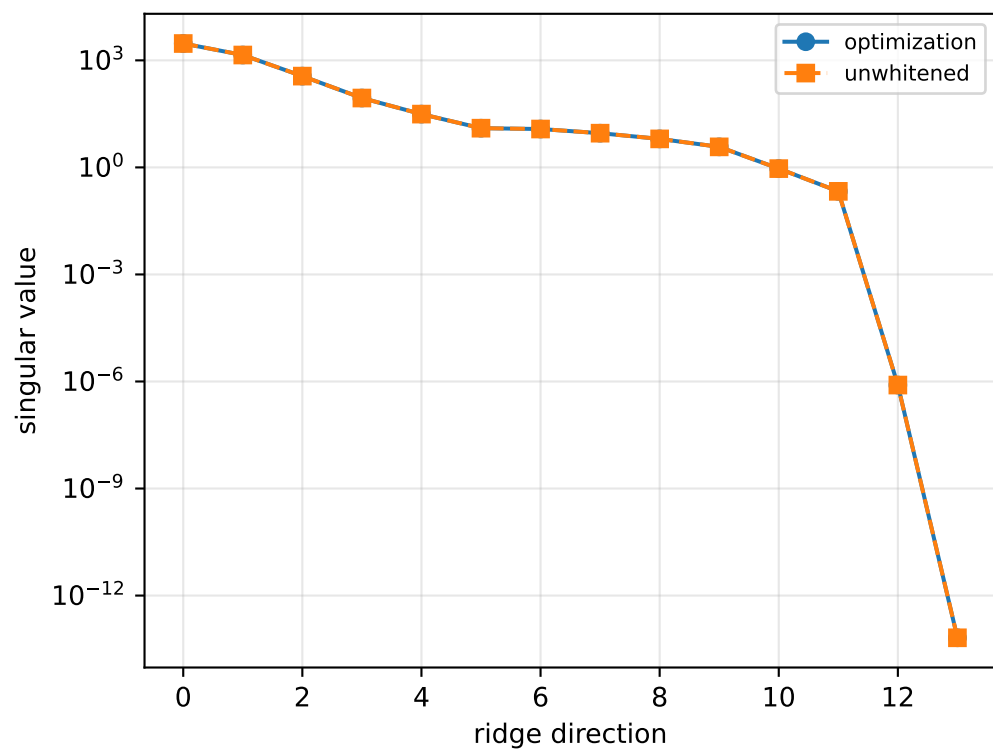
```
[[ 0.24099437 -0.00512007  0.01764571]
 [-0.00512007  0.1097261  -0.00398681]
 [ 0.01764571 -0.00398681  0.13761029]]
```

scale-free f/m: [0.44478472 0.43471687 0.22853732 0.27956522]

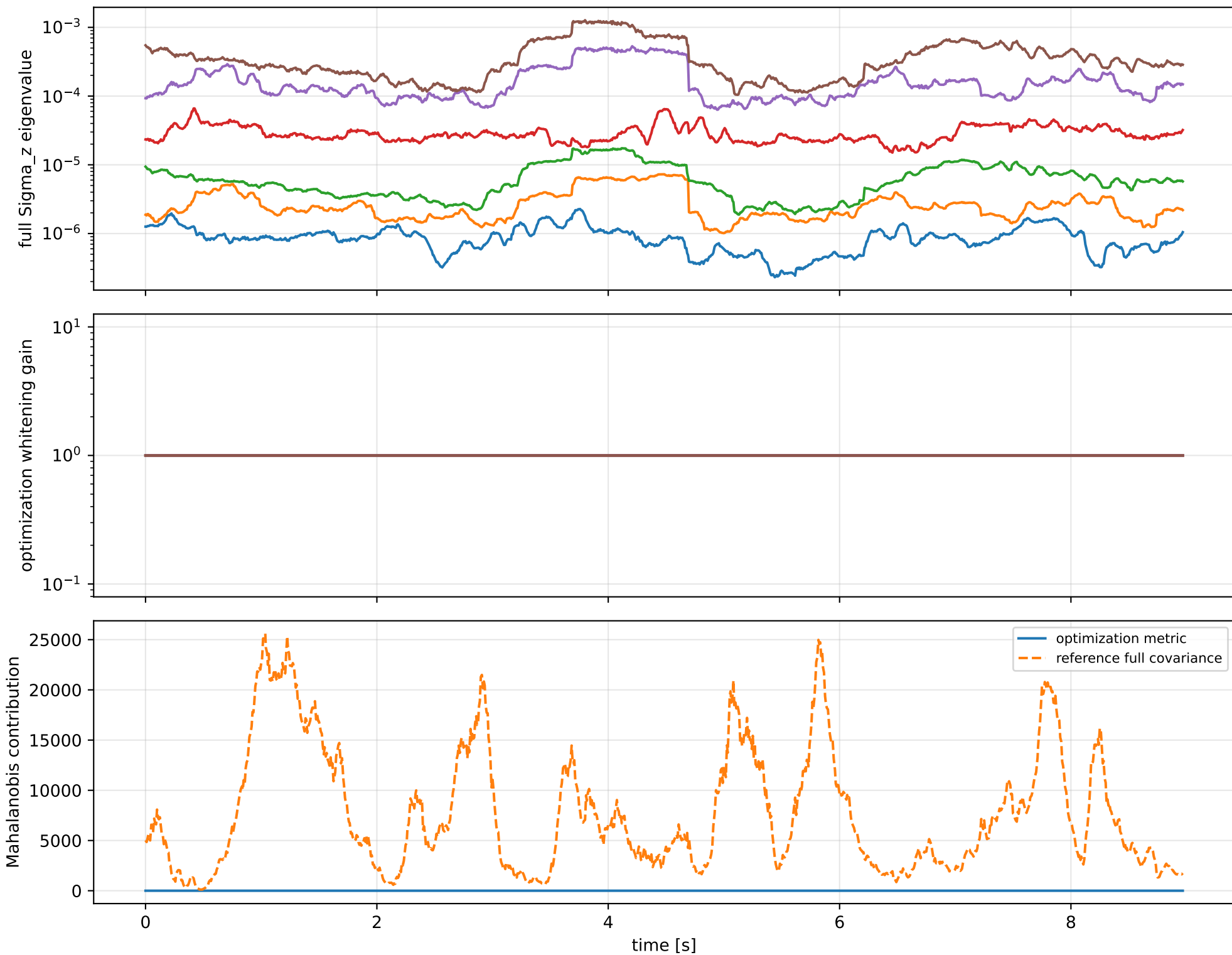
rotor lag cell: (0.175582647, 0.175611973] s

representative: 0.17559731 s

gimbal_measured_linear: ridge spectrum (rank 13, nullity 1, $||v||=7.47e-14$)



gimbal_measured_linear: optimization vs reference covariance



gimbal_measured_linear: wrench / closure (force max 1.48e-14, torque max 8.33e-17)

